

Tristan Ryoma Fuchs

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Master's student (M2)

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Arita Group · MERIT-WINGS program student

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RESEARCH INTERESTS

Superconductivity · first-principles (DFT) calculations

EDUCATION

- | | |
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| Apr 2025 – present | Master's Program in Physics, The University of Tokyo
Supervisor: Ryotaro Arita. |
| Apr 2021 – Mar 2025 | B.Sc. in Physics, Department of Physics, Faculty of Science, Tohoku University
Supervisor: Takashi Koretsune.
First-principles Migdal–Eliashberg calculation of the superconducting gap functions of Al and H ₃ S. |
| Sep 2016 – Jun 2020 | Shanghai Jincai High School, International Division, Shanghai, China
Grades 9–12; International Baccalaureate Diploma Programme (Grades 11–12). |
| Apr 2008 – Aug 2016 | Shanghai Japanese School, Pudong Campus, Shanghai, China
Grades 1–9. |

PUBLICATIONS

1. **T. R. Fuchs**, T. Nomoto, H. Watanabe, and R. Arita, “Microscopic calculation of coherence lengths and magnetic penetration depth in multiband superconductors,” [arXiv:2609.06014](#) [cond-mat.supr-con] (2026), under review.

PRESENTATIONS

International Conferences

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| Dec 2026 | Microscopic Calculation of Characteristic Length Scales in Multi-Band Superconductors POSTER SCHEDULED
T. R. Fuchs , T. Nomoto, H. Watanabe, R. Arita — International Quantum Materials Symposium (i-QMS 2026), Jeju, Korea |
| Nov 2026 | Coherence Lengths and Magnetic Penetration Depth in Multiband Superconductors from Microscopic Calculations POSTER SCHEDULED
T. R. Fuchs , T. Nomoto, H. Watanabe, R. Arita — International Workshop on Frontiers in Ab-initio Many-Body Interactions (FAMBI 2026), Sendai, Japan |
| Sep 2026 | Microscopic Calculation of Coherence Lengths and Magnetic Penetration Depth in Multi-Band Superconductors POSTER
T. R. Fuchs , T. Nomoto, H. Watanabe, R. Arita — International Conference on Asymmetric Quantum Matters (Asymmetry 2026), Osaka, Japan |
| Feb 2026 | Microscopic Calculation of Coherence Lengths and Magnetic Penetration Depth in Two Band Superconductors POSTER
T. R. Fuchs , T. Nomoto, H. Watanabe, R. Arita — CEMS Topical Meeting on Recent Trends in Superconductivity, RIKEN, Wako, Japan |
| Sep 2025 | Microscopic Calculation of Coherence Length and Magnetic Penetration Depth in Superconductors POSTER
T. R. Fuchs , T. Nomoto, H. Watanabe, R. Arita — Autumn School on Correlated Electrons: Understanding Correlated Materials with DMFT, Forschungszentrum Jülich, Germany |

Domestic Conferences (Japan)

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| Sep 2026 | Calculation of Coherence Lengths and Magnetic Penetration Depth Based on the Real-Space Bogoliubov–de Gennes Method POSTER
T. R. Fuchs , T. Nomoto, H. Watanabe, R. Arita — 81st Annual Meeting of the Physical Society of Japan (JPS), The University of Tokyo, Komaba Campus (in Japanese) |
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Sep 2026	Microscopic Calculation of Characteristic Length Scales in Multi-band Superconductors Based on the Extended Ginzburg–Landau Method POSTER BEST POSTER AWARD <u>T. R. Fuchs</u> , T. Nomoto, H. Watanabe, R. Arita — MERIT Camp 2026 (MERIT-WINGS, The University of Tokyo), Minano, Saitama
Aug 2026	Microscopic Calculation of Coherence Lengths and Magnetic Penetration Depth in Multi-Band Superconductors Based on Extended Ginzburg–Landau Theory ORAL POSTER <u>T. R. Fuchs</u> , T. Nomoto, H. Watanabe, R. Arita — 71st Condensed Matter Physics Summer School (物性若手夏の学校), Aso, Kumamoto (in Japanese)
May 2026	Microscopic Calculation of Coherence Lengths and Magnetic Penetration Depth in Multi-Band Superconductors POSTER <u>T. R. Fuchs</u> , T. Nomoto, H. Watanabe, R. Arita — Kickoff Meeting of the Transformative Research Area (A) “Asymmetric Quantum Matters,” Nasu, Tochigi (in Japanese)
Mar 2026	Microscopic Calculation of Coherence Lengths and Magnetic Penetration Depth in Two-Band Superconductors ORAL <u>T. R. Fuchs</u> , T. Nomoto, H. Watanabe, R. Arita — JPS 2026 Spring Meeting, online (in Japanese)
Mar 2026	Microscopic Calculation of Coherence Lengths and Magnetic Penetration Depth in Two-Band Superconductors POSTER BEST POSTER AWARD <u>T. R. Fuchs</u> , T. Nomoto, H. Watanabe, R. Arita — Computational Physics Spring School 2026 (計算物理春の学校), Okinawa (in Japanese)

AWARDS AND FELLOWSHIPS

Sep 2026	Best Poster Award, MERIT Camp 2026
Mar 2026	Best Poster Award, Computational Physics Spring School 2026 (計算物理春の学校)
Oct 2025 – present	MERIT-WINGS Program Student World-leading Innovative Graduate Study Program for Materials Research, Information, and Technology (MERIT-WINGS), The University of Tokyo

EXPERIENCE

Dec 2023 – present	Student Staff Lead, Nagase Brothers Inc. (education company) Oct 2024 – present: development of internal automation tools at Yotsuya Otsuka Dec 2023 – Mar 2026: project management at Toshin Digital University
Spring 2024	Teaching Assistant, 学問論 (first-year general education course), Tohoku University

LEADERSHIP AND ACTIVITIES

Oct 2025 – Sep 2026	Deputy Organizer, MERIT Camp — The University of Tokyo
Oct 2023 – Mar 2025	Student Staff, Kita-Aobayama Library (Science & Pharmacy Library) — Tohoku University
Apr 2022 – Mar 2023	Vice President, 物理の会 (“Butsuri no Kai,” physics seminar society) — Tohoku University

CERTIFICATIONS AND TEST SCORES

2026	EIKEN Grade 1 実用英語技能検定（英検）1級
2025	Applied Information Technology Engineer Examination 応用情報技術者試験 — passed
2024	TOEFL iBT 105
2024	TOEIC Listening & Reading 960
2022	HSK Level 4 汉语水平考试 (Chinese Proficiency Test)

LANGUAGES AND SKILLS

Languages: Japanese (native), English (fluent), German (conversational), Chinese (basic)

Programming: Julia (main language for research), Python, Google Apps Script